Arista Access Node KR-ACC-82 Hardware/Environmental Fault ALARM_1033 Warning SOP

Node: KR-ACC-82 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_1033

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for the ALARM_1033 warning triggered by hardware/environmental faults in the KR-ACC-82 Arista Access Node. This alarm typically indicates an ASIC memory parity error, line card crash, or power supply redundancy loss within the node. The underlying cause of these faults is attributed to the Arista 7250SX line card's ASIC memory parity checking mechanism, which detects and reports errors in the memory array. Additionally, the power supply redundancy feature is designed to ensure continuous operation of the node in case of a single power supply failure. However, if both power supplies fail, the node may experience a hardware failure, resulting in an ALARM_1033 warning.

Pre-requisite Checks

- Check: Verify that the KR Puram Access site location is accessible for remote hands to perform necessary interventions.

- Check: Confirm that the redundant link status is operational and can be used as a backup in case of a primary link failure.

Potential Root Causes

Cause	Probability
ASIC memory parity error due to faulty memory array	High
Line card crash due to software or hardware malfunction	Medium
Power supply redundancy loss due to failure of both power supplies	Low

Diagnostic Commands & Expected Outputs

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Command: show system internal bootvar, Expected Output: Boot variables for the system, Description: This command displays the boot variables for the system, including the boot device and the configuration file used during boot-up.

Command: show system internal memory error detail, Expected Output: Memory error details, Description: This command displays detailed information about memory errors, including the type of error, the location of the error, and the timestamp for the error.

Command: show system internal power supply detail, Expected Output: Power supply details, Description: This command displays detailed information about the power supplies, including the status of the
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power supplies, the current power consumption, and the redundancy
status.
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Resolution Steps

Step 1: Step: Power cycle the node to clear any transient errors, Command: reload, Description: This command restarts the node, clearing any transient errors and allowing the node to reinitialize the ASIC memory parity checking mechanism.

Step 2: Step: Replace the faulty line card with a new one, Command: no linecard 0/1/1, Description: This command removes the faulty line card from the node, and the system will automatically detect and add a new line card to replace it.

Step 3: Step: Replace both power supplies with new ones to ensure redundancy, Command: no power supply 0, Description: This command removes the faulty power supply from the node, and the system will automatically detect and add a new power supply to replace it.

Post-Check Verification

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Command: show system internal memory error detail, Expected Output: No
memory errors detected, Description: This command verifies that no
memory errors are detected after replacing the faulty line card or
power supplies.

Command: show system internal power supply detail, Expected Output:
Power supplies redundancy status is operational, Description: This
command verifies that the power supplies redundancy status is
operational after replacing both power supplies.
```

Roll-back Procedures

- Step: Restore the original configuration file if it was modified during the intervention, Command: configure replace bootflash:original_config.txt, Description: This command restores the original configuration file if it was modified during the intervention, ensuring that the node reverts to its original configuration.

- Step: Revert the node to its original state before the intervention, Command: reload discard, Description: This command reverts the node to its original state before the intervention, ensuring that the node reverts to its original configuration and state.